Predicting Patient Outcomes: A Comparison of Image-Based and Clinical Models using Radiomics

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Introduction

In this project, the business problem we are trying to solve is how we might design a multimodal predictive algorithm that can accurately predict cardiopulmonary disease, specifically pneumonia and pneumothorax, based on chest X-ray image data and hospital electronic health record (EHR) data using supervised and unsupervised machine learning methods. Our team was motivated to work on this because we all shared a common interest in healthcare applications of AI and are either working in the field currently or have future ambitions.

As the demand for healthcare services grows, there is also an increased utilization of imaging services such as X-rays. Combined with workforce shortages, this is putting strain on our health systems and clinicians. Our solution has the potential to alleviate the diagnostic burden and reduce turnaround time for diagnostic clinical workflows so that patients can get the right care at the right time. In addition, our model could have transfer learnings across other non-cardiopulmonary disease states. Another novel application of our work was that we used lower resolution JPEG versions of the radiology images instead of the original DICOM format, which can pose storage and computation challenges but have richer hidden features that are undetectable to the naked eye.

For the supervised portion of our project, we evaluated three supervised model families to predict pneumonia and pneumothorax: a convolutional neural network (CNN) based on a pretrained ResNet50 architecture trained on chest X-ray images, and two models trained on structured EHR features using logistic regression and tree-based XGBoost. The CNN used transfer learning with RadImageNet, standard image preprocessing, and weighted BCE loss to address class imbalance, with pre-defined training/validation/test labels and subject-wise five-fold cross-validation for evaluation. The structured EHR models were trained on aggregated lab features summarized across a clinical timeline, with regularization and hyperparameter tuning via cross-validation to ensure fair comparison between our nonlinear and linear approaches.

The main findings from our supervised work revealed that across both diseases, the image-based ResNet50 models outperformed the EHR-based models. For pneumonia, ResNet50 achieved the highest performance based on AUROC and AUPRC, exceeding logistic regression and XGBoost. For pneumothorax, it also performed better but XGBoost was a close second. Interpretability using SHAP and Captum showed that CNN focused on clinically relevant regions of chest X-rays, with pneumonia signals concentrated in the lower lung fields and pneumothorax signals were more prominent in the edges of the lungs and more spread out. Pneumothorax predictions were more stable to the clearer radiographic boundaries, where pneumonia presented more diffuse patterns.

During our unsupervised modeling, for each chest x-ray we extracted the 2048-dimensional image embeddings using a pretrained model specializing in radiology (ResNet50) and removed the final classification layer. We used the learned embeddings for our clustering algorithms instead of the raw image pixel value because they are compact (2048 dimensions vs. 6.7 million raw grayscale pixels),

encode semantic meaning and are efficient for storage and computation. We then applied Principal Component Analysis (PCA) to further reduce dimensionality down to the essential components. These PCA reduced image embeddings served as inputs to both clustering algorithms.

Similarly, the main findings from our unsupervised ML analysis showed that both k-means and DBSCAN identified some meaningful global structure in our image dataset (silhouette score ~0.15) but neither method produced clusters that aligned strongly with pneumonia labels (adjusted rand index ~ 0). This suggests that pneumonia does not correspond to a discrete separable cluster in the embedding space. Instead, pneumonia may occupy a continuum or gradient. We also observed that the clusters were influenced by global factors such as the viewing position for the chest xray (and presence of medical devices such as ECG leads).

Looking ahead, there is a lot of opportunity for future work to build multimodal predictive systems by combining structured EHR data and unstructured data to improve prediction of patient outcomes (e.g. Length of Stay and 30-day Mortality) that can impact downstream decisions on appropriate care escalations and efficient utilization of healthcare resources. These models could provide early signals of patient "ballooning events" where one's condition deteriorates fast and assist clinicians with prioritizing high-risk cases. We could also combine this with clinical LLMs to develop appropriate care pathways. With additional data, computational resources, and model refinement we think our framework could be extended to additional disease states and incorporate additional modalities of unstructured clinical data such as video, audio and clinical notes as well.

Related work

- **"Pneumonia detection in chest X-ray images using an ensemble of deep learning models"**
  This study used an ensemble of CNNs to detect pneumonia from chest X-Rays. Our project is different because we look at EHR data in addition to radiographic data. We also use RadImageNet which is not one of the CNNs utilized in the study by Kundu et al.
- **"Glaucoma diagnosis in the Chinese context: An uncertainty information-centric Bayesian deep learning model"**
  This study uses a single Bayesian learning model trained on combined image and text-based data to diagnose Glaucoma. Our project is different because we developed multiple unimodal models rather than a single multimodal model. We also focus on chest X-rays rather than eye imaging.
- **"Machine learning-based model for predicting all-cause mortality in severe pneumonia"**
  This study uses clinical data to predict mortality in patients with pneumonia. In contrast, our study focuses on predicting the presence of pneumonia rather than long-term effects. However, this study provides good insight into possible steps forward for future research.

Data Source(s)

Our team is taking advantage of the RadImageNet dataset, which is similar to the well-known ImageNet dataset but is specifically designed and curated around radiological images. This dataset was made available by MIT and requires compliance training before getting access to it.

ResNet50 was trained on the MIMIC Chest X-ray (MIMIC-CXR) database. It consists of 377,110 images from 227,835 radiographic studies from Beth Israel Deaconess Medical Center in Boston, MA (Johnson et al., 2024a). Specifically, the MIMIC-CXR-JPG was used in this project. It is derived from MIMIC-CXR but stores the images as JPG files instead of DICOM (Johnson et al., 2024b). DICOM is a binary file containing large amounts of metadata with pixel values, including pixel depths not perceivable by the

human eye. However, this creates issues of interpretability and storage size concerns. Johnson and colleagues (2024b) extracted image pixels from the DICOMs using Python's pydicom library, normalized from a range of zero to 255, and converted to JPGs using OpenCV. Three data sources were derived from the database and stored on the University of Michigan Great Lakes HPC Cluster: mimic-cxr-2.0.0-chexpert.csv.gz, mimic-cxr-2.0.0-split.csv.gz, and images within individual folders. For reference, a typical DICOM image comparison with JPG is provided in Appendix A.

The chexpert CSV contains labels derived from the open-source labeling tool CheXpert (Johnson et al., 2024b). Variables include subject_id, unique integer values for each patient, study_id, an integer unique to each radiology report with one or more images associated with it, with the remaining columns labeled for a specific diagnosis or as "no finding". The diagnosis columns include 1.0, 0.0, -1.0, or missing (empty) labels, defined as positive, negative, uncertain, or not mentioned respectively.

The split CSV provides a reference dataset split for studies using MIMIC-CXR-JPG (Johnson et al., 2024b). The file includes the same subject_id and study_id columns as chexpert used for merging patients and their radiology reports across datasets. The split column is a string indicating how the data is split between 'train', 'validate', and 'test' sets.

The "files" folder contains the directory of chest x-ray images. Within it is a folder for each patient containing folders for each radiology report, with each report containing one or more JPG images. The patient and report folders were used to replicate the subject_id and study_id integers for merging with the chexpert and split files. An image_path column was created with file paths leading to each image in our Great Lakes directory. After merging, 69,580 pneumonia images and 71,976 pneumothorax images labeled as positive or negative were included for analysis.

In addition to imaging data, structured Electronic Health Record (EHR) data was obtained from the MIMIC-IV database. MIMIC-IV is distributed through PhysioNet as compressed CSV tables containing clinical information such as laboratory measurements, demographic information, procedures, and medications. For the EHR-based models in this study, laboratory test events were extracted from the labevents table and linked to descriptive laboratory names using the d_labitems dictionary table. Demographic information including anchor_age and gender were also included. Laboratory tests used in the models included blood urea nitrogen, creatinine, sodium, potassium, magnesium, bicarbonate, and hematocrit, among others. In total, MIMIC-IV included data on 364,627 patients and 546,028 admissions.

Feature engineering

CheXpert reports which coded pneumonia or pneumothorax as 0.0 or 1.0 were filtered for. Standard ImageNet normalization was performed on all images. This produced a dataset with subject_id, study_id, image_path, the target variable (pneumonia or pneumothorax), and the split column.

For the EHR models, laboratory measurements from the MIMIC-IV dataset were used as predictors for disease classification. Several preprocessing and feature engineering steps were required to transform the data into a suitable tabular format.

First, Chest X-ray studies from MIMIC-CXR-JPG were linked to hospital encounters in MIMIC-IV through shared identifiers (subject_id, study_id, and hadm_id). For each study, lab values recorded within the 24 hours preceding the imaging were extracted. Restricting features to measurements recorded before the imaging ensured that the models were comparable and avoided leakage from future clinical events. Since multiple measurements sometimes occurred within this 24-hour window, values for each test were

aggregated using maximum, minimum, and mean. These summary statistics represented both the average and extreme states of each patient. To preserve the full dataset, missing values were not removed and were instead handled during model training. Logistic regression models used median imputation for missing values, while XGBoost handled missing values through learned split directions within the decision trees.

After aggregation and preprocessing, each row of the final dataset represented a single imaging study paired with its associated laboratory summaries from the preceding 24 hours. These engineered features served as inputs for the logistic regression and XGBoost supervised learning models. A complete list of the final laboratory features used in the EHR models is provided in Appendix B.

Supervised Learning

Our supervised models included image-based convolutional neural networks (CNNs) in addition to a logistic regression and XGBoost models trained on structured EHR features. Together, these models represent 3 diverse model families that provide a useful baseline for comparison between image-based and EHR-based approaches.

CNNs have wide application in diagnosis through radiological images (Hossain et al., 2022). However, medical imaging databases are more scarce due to the difficulty of curating and anonymizing them. A common solution is transfer learning, whereby a model trained for one deep learning task can be deployed for a related task (Mei et al., 2022). Additionally, these pre-trained models can save computational sources versus those developed from scratch. One class of pre-trained models include those trained on ImageNet, an open source dataset of millions of natural world images, including ResNet50 trained on over one million ImageNet images (Ikechukwu et al., 2021; Mei et al, 2022). ResNet50 consists of 50 layers, including convolutional blocks with average pooling and a softmax layer used for classification.

Subject-wise five-fold cross-validation was performed for pneumonia and pneumothorax. Using MIMIC-CXR-JPG reference splits yielded a training and validation pneumonia prevalence of 37.5% and a test prevalence of 52.4%. For pneumothorax, training and validation prevalence was 19.9% and test prevalence was 12.7%. For both diseases, the same baseline ResNet50 model was trained and the same edits made in versions two and three. This was done to test the strength of the model and effects of different adjustments under varying diseases and class composition scenarios.

The baseline ResNet50 models were implemented in ResNet50_pneumonia_v1.ipynb and ResNet50_pneumothorax_v1.ipynb. They were trained using default weights. Standard ImageNet normalization of images was performed during preprocessing. BCEWithLogitsLoss loss function weighted towards the positive class was used. An Adam optimizer was applied to all model parameters with a learning rate of 1e-4.

Some overfitting was seen in the area under the precision-recall curve (AUPRC) and area under the receiver operating characteristic curve (AUROC) by the third epoch. Partial layer freezing was done as a regularization technique to try and retain low-level features, while concentrating on high-level, domain-specific features in later layers (Luthfiarta et al, 2025). Each model was frozen except for the last residual block (layer 4) and the fully connected (FC) classifier with the Adam optimizer set to only work on unfrozen parameters.

Then freezing was removed since performance did not improve to retain a full network, while dropout was added to the FC and weight decay to the Adam optimizer to try to reduce overfitting. While performance was slightly better than the frozen models, the baseline models still performed the best, suggesting aggressive, full-model fine-tuning is needed.

Logistic regression was implemented as a probabilistic linear classifier that models the log-odds of disease as a linear combination of input features. Logistic regression is widely used in clinical research because many medical prediction tasks involve binary outcomes such as the presence or absence of a disease (Zabor et al., 2021). In this study, laboratory measurements were aggregated within a fixed clinical time window and summarized using statistical features such as mean, minimum, and maximum values. These features formed a structured tabular dataset that could be directly used by linear models. Regularization was applied to prevent overfitting, and hyperparameters such as the regularization strength were explored using cross-validation.

To capture non-linear relationships between input variables and disease outcomes, we also trained gradient-boosted decision tree models using XGBoost. XGBoost is a tree-based method that was well-suited for our task because it can handle non-linear relationships between predictors and does not require feature scaling. XGBoost has been widely adopted across industries including healthcare and has been used successfully with EHR data before (Badhoutiya et al., 2023). Hyperparameter tuning was performed using randomized search with cross-validation across parameters such as learning rate and tree depth.

Both logistic regression and XGBoost models were trained using the same structured EHR feature representation to ensure comparability across model families. This allowed us to assess whether nonlinear tree-based models provided meaningful improvements over linear probabilistic models when predicting the presence of pneumonia and pneumothorax.

Supervised Evaluation

Overall results reporting

Model performance was evaluated using both AUROC and AUPRC. AUROC measures how well a model ranks positive cases above negative cases across all thresholds, thus considering both true positives and false positives (Zhang, 2025). AUPRC measures a model's trade-off between precision and recall across thresholds (Wen et al., 2024). The use of AUROC is appropriate for more balanced datasets, while AUPRC is for imbalanced datasets, covering both scenarios seen in pneumonia and pneumothorax. AUROC can gauge separation of healthy patients from unhealthy ones, while AUPRC shows how many at-risk patients can be flagged while considering the cost of screening healthy ones. Thus, using both measures provides a holistic view of our model while considering different needs of health care providers.

In Table 1, we report the mean metric across 5 cross-validation folds for our best-performing models from each family for both pneumonia and pneumothorax as required to avoid overinterpreting a single split.

Disease	Model Family	AUROC (mean ± sd)	AUPRC (mean ± sd)
Pneumonia	Logistic Regression	0.609 ± 0.006	0.554 ± 0.007
Pneumonia	XGBoost	0.625 ± 0.012	0.579 ± 0.012

Pneumonia	CNN (ResNet50)	0.779 ± 0.006	0.696 ± 0.012
Pneumothorax	Logistic Regression	0.618 ± 0.007	0.285 ± 0.006
Pneumothorax	XGBoost	0.722 ± 0.006	0.479 ± 0.005
Pneumothorax	CNN (ResNet50)	0.793 ± 0.005	0.564 ± 0.016

Table 1: Results table comparing logistic, XGBoost, and ResNet50 across pathologies. ResNet50 performed the best in both cases.

The image-based ResNet50 models outperformed the EHR models (XGBoost and Logistic Regression) for both diseases. The performance gap is especially pronounced for pneumonia, which is consistent with the fact that pneumonia is primarily diagnosed radiographically. The XGBoost model for pneumothorax does achieve an AUROC score that approaches the AUROC score of the CNN model for pneumothorax. This indicates that a multimodal model could improve performance in future work.

In-Depth Evaluation

An in-depth evaluation of the best-performing pneumonia and pneumothorax models, ResNet50, was conducted to assess the strengths and weaknesses of the model. This included: feature importance, ablation analysis, sensitivity analysis, and identification of tradeoffs[1].

Feature Importance and Ablation Analysis

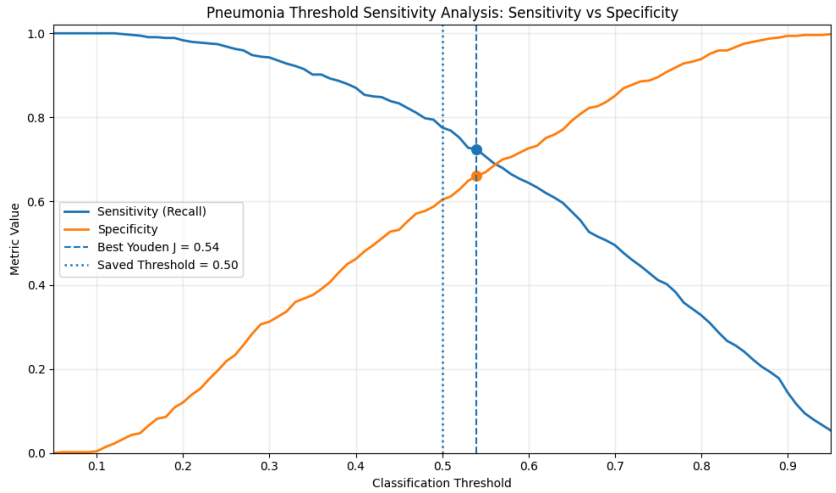
Feature importance was performed using the SHapley Additive exPlanations (SHAP) library. SHAP image plotting displays images alongside the class(es) of your model highlighting in red areas which causes an image to be classified to that class (Lundberg, 2018). Appendix D shows SHAP output for pneumonia and pneumothorax. Pneumonia's feature importance tends to be more concentrated in the bottom half of chest x-rays. Meanwhile, pneumothorax's feature importance shows more spread across chest x-rays, with the top left and right corners of chests playing a larger role in classification.

Ablation analysis was performed using the Captum library. This involved replacing input features, in this case image patches, with a baseline, rerunning the model, and seeing how the results change. Baselines are denoted with black pixels (Appendix E). Red patches mean the patch supports disease identification, blue patches mean the patch suppresses disease identification, and whiter colors mean the patch doesn't matter much. Pneumonia red was more diffuse, moderate edge sensitivity, and lower half emphasis. Pneumothorax red was more focal, showed strong edge sensitivity, and less lower half emphasis.

Sensitivity Analysis

We analyzed the sensitivity of the classification threshold. The model detects roughly 72% of pneumonia cases and 66% of non-pneumonia cases. Conversely, the model identifies roughly 74% of pneumothorax cases and correctly rejects roughly 74% of non-pneumothorax cases (Figure 1). These results suggest our model's confidence in classifying pneumothorax is higher than pneumonia. If threshold were shifted for pneumonia then sensitivity or specificity could fall or rise quickly (Figure 2). This could be due to pneumonia having fuzzier radiographic patterns, producing overlapping probabilities, and diffused pathology. Meanwhile, pneumothorax could have sharper boundaries, more stable probabilities, and clearer pathology.

[1] For an additional in-depth evaluation of the best performing EHR model please refer to Appendix C.



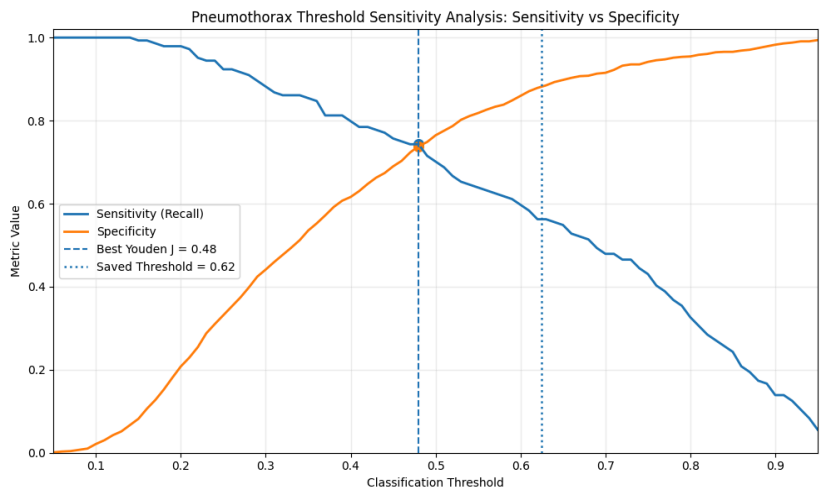


Figure 1: Threshold Sensitivity Analysis for Pneumonia & Pneumothorax
Source: ResNet50_ThresholdSensitivity_Pneumonia.ipynb &
ResNet50_ThresholdSensitivity_Pneumothorax.ipynb

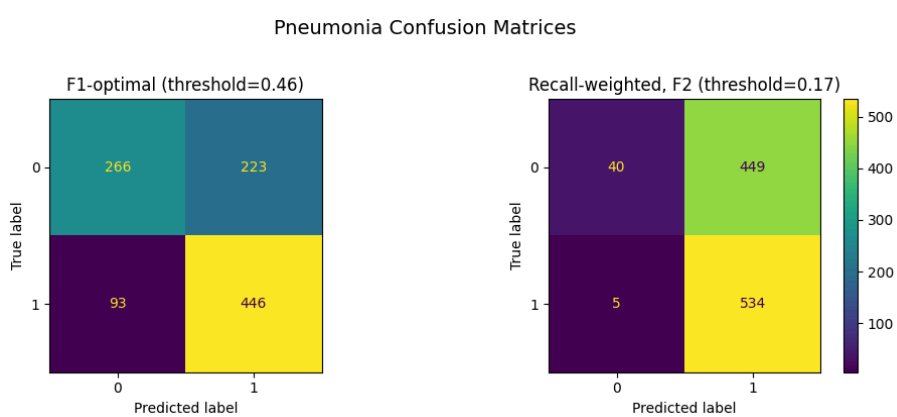


Figure 2: Confusion Matrix F1-Based versus Recall-Based
Source: plots_only_pneumonia.ipynb

Identify Tradeoffs

Given the results of our threshold sensitivity analysis, precision-recall tradeoffs differ between pathologies. Pneumothorax can support a higher recall threshold without totally sacrificing precision. However, choosing a higher recall threshold for pneumonia brings higher risks of false-positives, which can burden health care providers.

Changes to modeling architecture or data type can be considered for fuzzier pneumonia diagnosis. ResNet101 would introduce more layered CNNs, potentially lift pneumonia AUROC, while retaining DICOM images may uncover subtle patterns. Though both of these changes could introduce computing challenges to hospitals.

Failure analysis

For each disease, we selected a high-confidence, false positive; a high-confidence, false negative; and a borderline case (Appendix F). Some common themes emerge. Medical devices like tubes and wiring could be introducing unfamiliar shapes. The portable AP films could denote a different X-ray acquisition strategy which confuses the model. Additionally, milder disease cases could provide smaller signal(s) and resemble normal variation. Future preprocessing segment medical devices so that the model can focus on lung tissue. Datasets could also have a greater balance of AP films versus other X-ray modalities and images from ICUs versus outpatients. Models can also be retrained with an emphasis on subtle positive cases and Captum/SHAP analyses can be re-run on subtle cases to confirm the model is paying attention to these.

Part B. Unsupervised Learning

Methods Description

For the unsupervised portion of the project, our team used the same MIMIC-CXR-JPG dataset as mentioned above (See Data Sources section).

We began the unsupervised portion with a set of key questions:

- Do unsupervised clusters derived from image embeddings align with known clinical labels for pneumonia versus non-pneumonia? Are the clusters driven by non-disease-related artifacts e.g. presence of medical devices during imaging study such as ECG leads, pacemakers, demographic differences (adult vs. child), etc.?
- Does a centroid-based method such as k-means produce clusters that meaningfully separate pneumonia cases or will a density-based method DBSCAN reveal more advanced structures?
- Do pneumonia cases form a discrete cluster or do they lie along a continuous gradient within the embedding manifold?
- To what extent do image artifacts e.g. medical devices such as ECG leads, pacemakers, etc. influence clustering structure?

Those questions along with past work in radiomics (Mei 2022, Remzan 2024) helped inform our unsupervised machine learning pipeline phases below:

- **Phase 0: Data Exploration, Validation & Pre-processing**
    - Merge MIMIC-CXR-JPG metadata and CheXpert labels; filter on pneumonia labels (0/1).
    - Preprocess images using standard ImageNet normalization.
    - Extract 2048-d embeddings from pre-trained ResNet50 model (Lab-rasool, 2021).

    - Standardize embeddings using and dimensionality reduction using PCA
    - Visualize top 50 principal components and PC1-PC2 by pneumonia label.
    - Use PCA-reduced embeddings for k-means, DBSCAN and geometric analysis.
- **Phase 1: k-means Clustering, Viz & Evaluation**
    - Sweep across a range of k = 2-10 on PCA embeddings.
    - Evaluate: silhouette, adjusted rand index (ARI), normalized mutual information (NMI).
    - Visualize via tSNE.
- **Phase 2: DBSCAN Clustering, Viz & Evaluation**
    - Optimize parameters for DBSCAN: eps (using k-distance plot) and range of min-samples.
    - Assess the number of clusters and noise labels
    - Evaluate: silhouette, adjusted rand index (ARI), normalized mutual information (NMI).
    - Visualize via tSNE.
- **Phase 3 (Future Work): Pneumonia Phenotypes**
    - Restrict Pneumonia flag = 1 and repeat the pipeline above: PCA, k-means, DBSCAN, evaluation and visualization.
    - Explore deeper structures within pneumonia to identify discrete subtype clusters.

Method Descriptions, Justifications and Model Hyperparameter Tuning

Given the high dimensionality of the extracted embeddings, we used Principal Component Analysis (PCA) for dimension reduction before proceeding with clustering analysis. We evaluated the expected variance ratio (EVR) across the first 50 principal components (PCs). Visualization of the first two principal components (Figure 3) revealed substantial overlap between pneumonia and non-pneumonia cases. Although the embedding space exhibits a clear global structure, the two classes are highly intermixed. This suggests that the top two PCs are not strongly associated with pneumonia status. This is consistent with our findings in the EVR analysis that shows that the first two PCs explained a modest 11.3% of the variance, whereas all 50 PC's explained 42.3% of the variance.

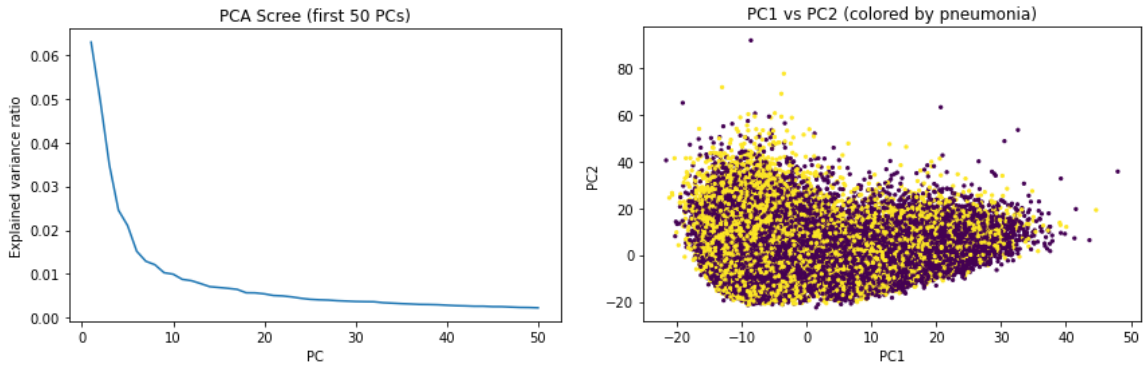


Figure 3: PCA Evaluation for first 50 principal components, and top 2 PC's

Next, we used those PCA-reduced embeddings with our two clustering methods. First we used k-means clustering, which is one of the simplest and most commonly used clustering algorithms. Given the large volume of data, high dimensionality and computational demands, k-means provided us with an efficient starting point. This method also allowed us to explore whether centroid based spherical clusters would align with the pneumonia dataset.

A key input parameter to tune for k-means is the number of clusters (k) we expect to see. Ideally, we would expect two clusters, one for each pneumonia label in our dataset. However, there could be other underlying geometric structures that represent more than just the disease state, which may allow for a

greater number of clusters so we explored a range of k values (2-10). The best k value was determined by using an evaluation scorecard to measure internal cluster structure cohesiveness using Silhouette score and external cluster validation with ground truth pneumonia labels using adjusted rand index (ARI) and normalized mutual information (NMI). See sensitivity analysis section for more details on the best k.

The other clustering method we used was DBSCAN (density-based spatial clustering of applications with noise), which unlike k-means doesn't require us to input a predefined number of clusters and can discover them on its own. Another advantage is that it can identify complex cluster shapes i.e. non-spherical, and also identify data points that are considered noise and not part of any given cluster. While it is slower than k-means, it was able to scale to our high-dimensional dataset.

For DBSCAN, there are two hyperparameters that can be used to tune the clusters performed: *eps* and *min_samples*. *Eps* is the distance from the cluster center that the algorithm looks to identify additional data points to be a part of that cluster and *min_samples* is the minimum number of data points required for those collection of data points to be considered a cluster. DBSCAN can be very sensitive to the parameter values, especially eps (Müller and Guido, 2017). See sensitivity analysis section for more details on the best eps and min_samples.

Unsupervised Evaluation

The unsupervised analysis using k-means revealed that having 3 clusters demonstrated the best Silhouette score (~0.15) but almost zero alignment with pneumonia labels (ARI and NMI ~ 0.01). This reveals that the clusters identified by those two methods were geometrically non-trivial but had almost zero association with pneumonia labels. When we visualize representative samples closest to those 3 cluster centroids, we can see that our original key question gets answered that there are more global artifacts that are associated with clusters instead of the pneumonia disease (figure 4). The top cluster shows images that have medical devices in the image e.g. ECG leads. The second cluster reveals that it is an anterior/posterior view position for the chest x-ray. Finally, the last cluster of images is made up of a lateral viewing position.

In addition, our team applied tSNE to a sample of the data to visualize the manifolds in the high dimensional space and overlaid the k-means clusters and DBSCAN clusters. We chose to apply it to a sample to save time and compute. We observed some global structure with a clear separation: one large elongated cluster on the left and another more compact cluster on the right. Since the first two k-means clusters had the same viewing angle, we can see that they are part of the elongated cluster, whereas the lateral view position is the smaller more compact cluster on the right. However, when we overlay the pneumonia labels on top of those clusters (Figure 5), we can see that there are no clean boundaries for pneumonia across those two clusters as it is diffusely spread across both. Lastly, when we used DBSCAN cluster analysis to identify noise points, we can see that there are

Cluster 0: Closest Images to Centroid

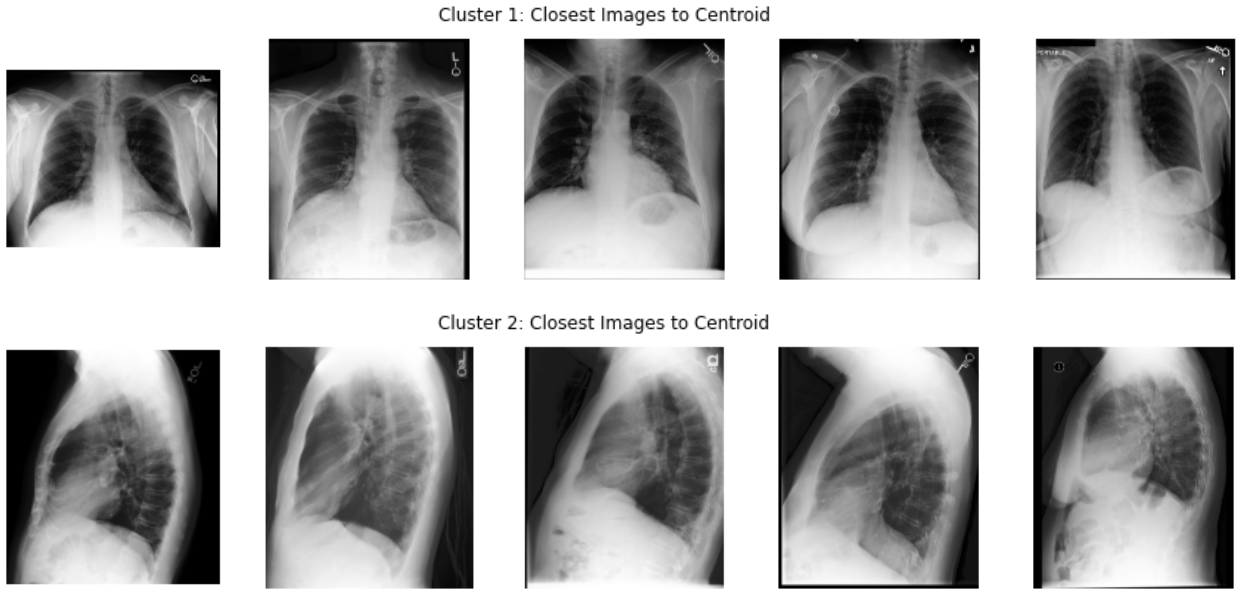


Figure 4: Sample images closest to the k-means cluster centroids

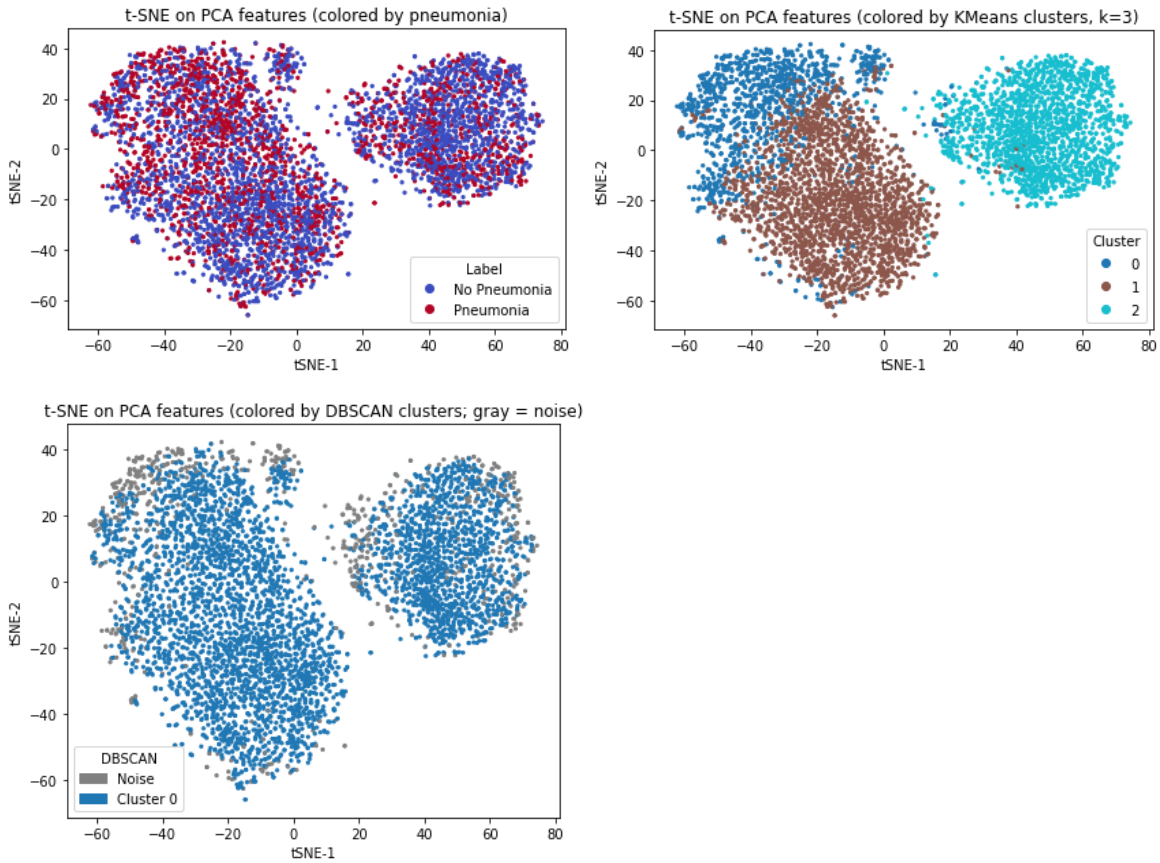


Figure 5: tSNE manifolds with pneumonia labels, k-means clusters and DBSCAN clusters

Overall results reporting

Sensitivity Analysis for k-means and DBSCAN

To arrive at the right number of clusters for k-means (best k = 3), we compared a range of k values against the evaluation scorecard of Silhouette score, ARI and NMI. While we calculated the Silhouette score for the entire dataset, we chose to limit calculations of ARI and NMI to a sample of the data points to save time. Our goal was to find the number of clusters k with the highest Silhouette score followed by a high ARI or NMI score. As you can see in Table 2, after k=3, the Silhouette score drops dramatically. This can be visualized in Figure 6 as well.

k	Silhouette Score	ARI sample	NMI sample	Min cluster	Max cluster	Sample size
2	0.148164	-0.005684	0.006998	24383	45197	5000
3	0.124773	0.014992	0.010791	13955	31777	5000
4	0.070416	0.012896	0.010994	10695	23396	5000
5	0.077715	0.004498	0.011970	7660	21671	5000
6	0.055150	0.009835	0.013602	6199	16673	5000
7	0.052997	0.008938	0.013136	5979	16322	5000
8	0.049328	0.008748	0.012403	3812	15101	5000
9	0.050078	0.010631	0.015334	3438	14110	5000
10	0.049656	0.009648	0.015206	3366	14028	5000

Table 2: Evaluation scorecard for cluster sizes for k-means

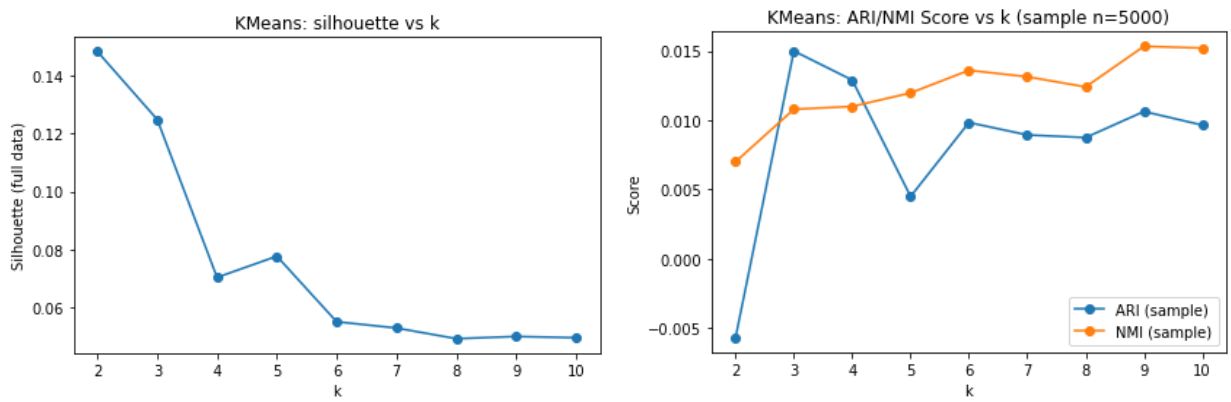


Figure 6: Silhouette score and ARI/NMI for k cluster range

Similarly, for DBSCAN we used the 'elbow/knee' method by utilizing a k-distance plot to identify the optimal value for eps (Ester 1996). This is a useful heuristic to help find good candidates for eps when the distances to the 10th nearest neighbors (minPts =10) tends to jump i.e. elbow/knee point. In our case, the optimal eps was 23.3. As you can see in the graph below, that's right when the line makes a sharp inflection upwards, meaning that clusters are shifting from denser areas to more sparse points, which is the underlying mechanism for DBSCAN clustering. The detailed results can be seen in Appendix G and are visualized in Figure 7 below.

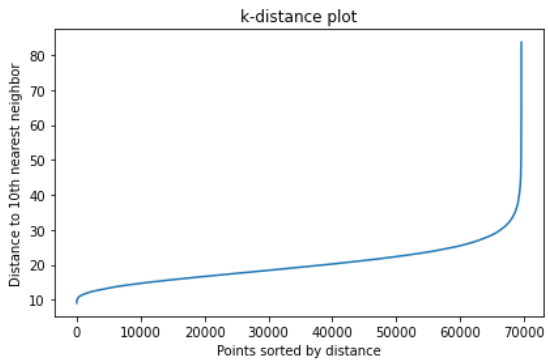


Figure 7: k-distance plot for DBSCAN elbow/knee method.

Discussion

For the supervised portion, we observed that while pneumonia achieved higher AUROC and AUPRC in ResNet50, the model showed greater confidence in predicting pneumothorax. Two factors were relevant. First, class imbalance was present in pneumothorax, while pneumonia was more balanced, both of which can bias AUROC and AUPRC. Second, pneumothorax showed more distinguishable features in the in-depth evaluation. This shows the importance of thoroughly assessing models beyond one or two evaluation metrics to guide future model improvements.

Running the ResNet50 was time-intensive, which required adjusting our model notebooks to save model checkpoints and results in the form of .pth and CSV files. This allowed for the most important artifacts to be saved so that later analysis of the models could be done efficiently. For example, ResNet50_Ablation.ipynb can be run using a model checkpoint .pth from a given fold and a model's test predictions CSV to produce Captum visuals without rerunning ResNet50.

There are several ways our solution could be extended with more time and resources. First, we would like to develop a multimodal model that combines both imaging data and EHR features. Integrating both modalities could improve predictive performance compared to either data source alone. Second, we would extend the task beyond disease diagnosis to predicting longer-term patient outcomes such as length of stay in hospital and mortality. These predictions would be extremely useful for care planning and risk assessment. Finally, there are opportunities to further improve our existing models. Potential improvements include using DICOM files directly rather than relying on converted JPG files and exploring additional pre-trained CNN models focused on medical imaging.

From completing Part B, we learned how unsupervised learning can be used to explore the geometric structure of high-dimensional image embeddings. Using embeddings extracted from a pretrained ResNet50 model, we applied PCA and clustering methods to investigate whether pneumonia cases formed natural groupings in the embedding space. One surprising result was that k-means clustering picked up on global structures in the chest x-ray data instead of the presence of pneumonia, including medical devices in the images (e.g. ECG leads) and viewing positions. For future work, it could be possible to train an ensemble of unsupervised models, one for each type of view position including one for the presence of medical devices on the body. While clustering revealed meaningful geometric structure, it did not align strongly with pneumonia labels thus suggesting that the dominant variation in the embeddings was due to global artifacts rather than disease status. A key challenge was the computational complexity of working with large, high-dimensional image datasets and sharing computing

resources on Great Lakes HPC. We addressed this by reducing dimensionality with PCA and sampling subsets of the data for visualization and evaluation.

With more time and resources, we would extend this work by identifying potential phenotypes within pneumonia-positive images and correlating these patterns with clinical severity measures from hospital EHR data. Our current analysis focuses primarily on geometric structure rather than clinical validation. We would also expand the evaluation framework to include additional clustering metrics such as the Davies–Bouldin score (lower is better) and the Calinski–Harabasz index (higher is better) to provide a more comprehensive assessment of cluster quality if we end up exploring a greater number of clusters.

Ethical Considerations

One ethical concern when deploying supervised models for medical diagnosis is the risk of incorrect predictions influencing clinical decisions. False negatives could delay treatment, and false positives could cause unnecessary testing and stress on both staff and patients. Another concern is potential bias in training data. Our training data may not represent all populations or could reflect bias in the treatment provided to people of different demographics. To address these issues, our models should only be used as a support tool and primary clinicians should remain responsible for final diagnoses. Additionally, the models should be evaluated across demographic subgroups to ensure consistent, unbiased behavior.

For our unsupervised analysis, the main ethical concern involves interpreting our clusters. It is possible that unsupervised clustering algorithms will group patients based on factors unrelated to disease such as imaging equipment or demographic differences. To mitigate this risk, clusters should be interpreted cautiously and validated using additional studies.

Statement of Work

- All: data pre-processing, project planning, assisting others with any issues
- Daniel: Supervised: EHR-based models (XGBoost and Logistic Regression). In-depth evaluation and failure analysis of XGBoost model.
- Ali: Unsupervised learning models (k-means, DBSCAN, tSNE), PCA, evaluation and visualization
- Samuel: Supervised: CNN-based imaging model ResNet50. Evaluation, metrics, and visuals for the model.

GenAI Disclosure

During the preparation of this work, the authors used GenAI tools for brainstorming and programming assistance. After using these tools, the authors reviewed and edited the content to align with project goals.

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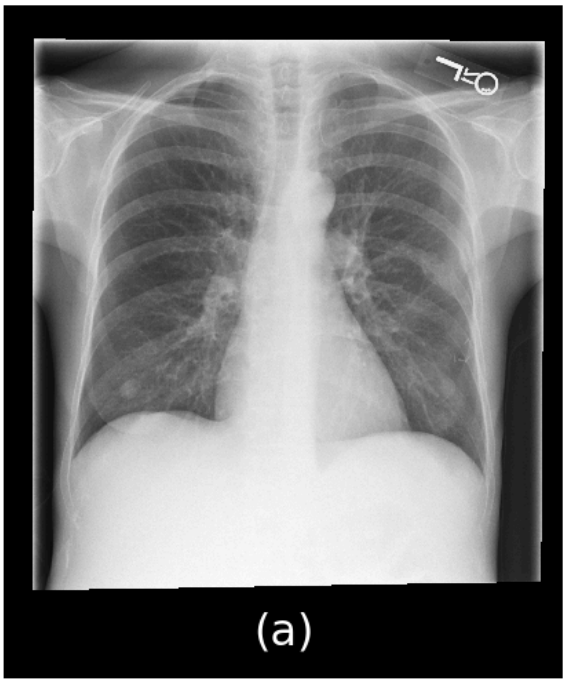
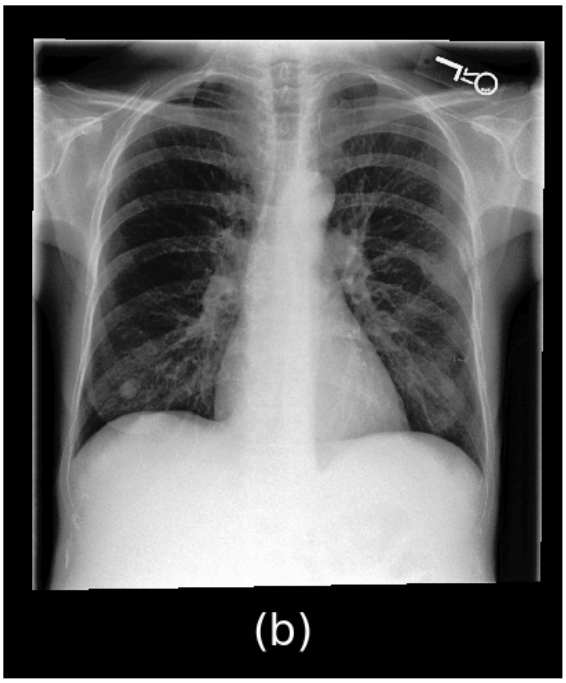
Appendix A: DICOM and JPG Image Comparison

Source:
https://github.com/MIT-LCP/mimic-cxr/blob/master/notebooks/compare-dicom-with-jpg.ipynb

A: DICOM, B: JPG





Appendix B: List of Variables used in EHR Model

ItemID	Laboratory test	Statistics included
50868	Anion Gap	max, mean, min
50882	Bicarbonate	max, mean, min
50902	Chloride	max, mean, min
50912	Creatinine	max, mean, min
50931	Glucose	max, mean, min
51221	Hematocrit	max, mean, min
50960	Magnesium	max, mean, min
50971	Potassium	max, mean, min
50983	Sodium	max, mean, min
51006	Urea Nitrogen	max, mean, min
—	anchor_age	demographic
—	gender	demographic

Appendix C: In-Depth Evaluation of Best-Performing EHR-Only Model

In-Depth Evaluation

Deeper analyses were conducted on our best-performing EHR-only model (XGBoost, tuned via randomized cross-validation), which was trained on data from MIMIC-IV to classify pneumonia and achieved a test AUPRC of 0.567 and AUROC of 0.623 on a held-out test set. The related code can be found in evaluate_xgb_pneumonia_best.py. In the following sections, we will refer to this model as best_EHR.

Feature Importance and Ablation Analysis

To understand which variables drove predictive performance for best_EHR, we conducted both permutation feature importance analysis and leave-one-feature-out (LOFO) ablation analysis on the test set.

Permutation importance is “model inspection technique that measures the contribution of each feature to a fitted model’s statistical performance on a given tabular dataset” (Scikit-Learn documentation). The method measures the degradation in model performance when the values of a feature are randomly shuffled. We used mean drop in test AURPC as our evaluation metric.

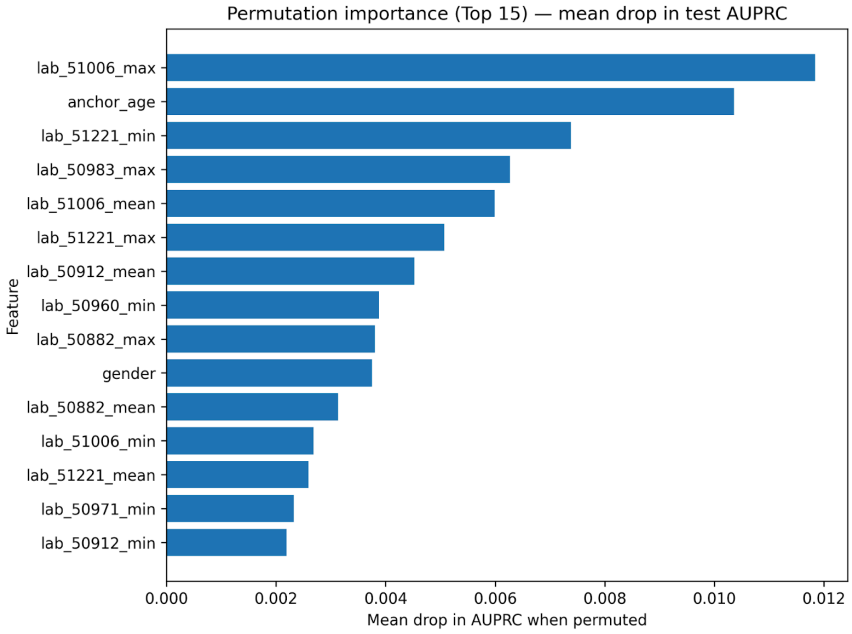


Figure 1: Permutation Importance

Figure 1 shows the top 15 features ranked by AURPC drop when permuted. The largest decrease was observed for lab_51006_max, which refers to Urea Nitrogen in blood. Since there are studies showing that blood urea nitrogen can be a predictive factor for pneumonia (Milas et al., 2022), this result is clinically plausible. Most of the top 15 variables are lab results, including creatinine, sodium, potassium, magnesium, bicarbonate, and hematocrit in blood. However, it should be noted that age and gender are two demographic variables that meaningfully contribute to model performance. Demographic variables require additional examination as discussed in our ethical considerations section. Importantly, the mean drop in AUPRC did not exceed 0.013 when permutating any single feature. This indicates that no single feature dominated our predictions.

Additionally, we conducted LOFO ablation analysis. For each feature, we removed it from the feature set, retrained the model, and evaluated test AURPC.

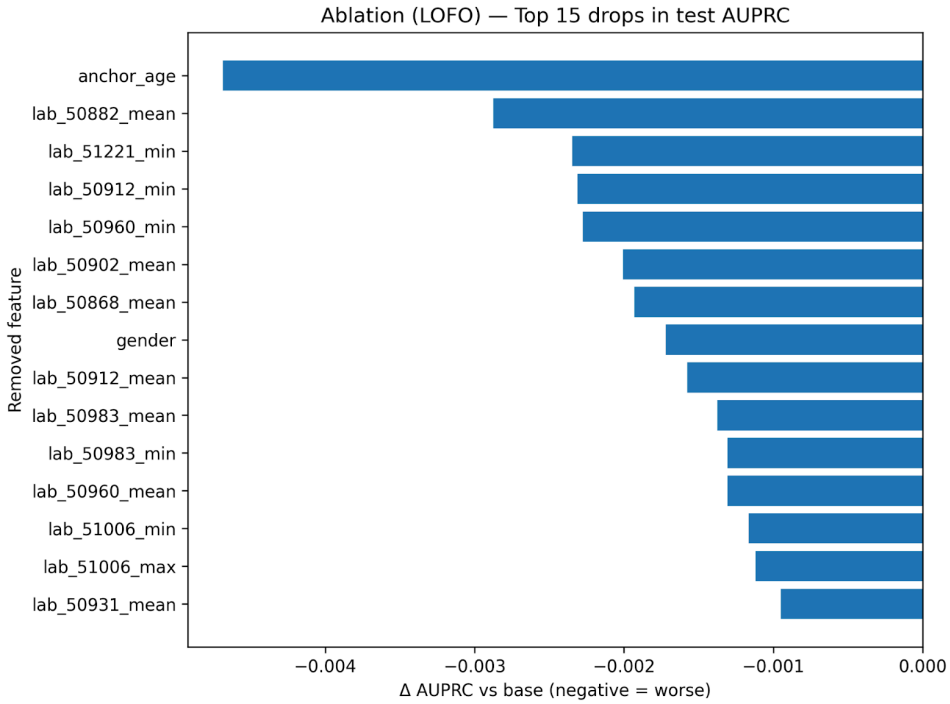


Figure 2: Leave One Feature Out Ablation

Figure 2 shows the largest decreases in AURPC when a feature is removed. The largest performance drop occurred when removing anchor_age (the patient's approximate age when admitted). People in certain age groups are at increased risk of getting pneumonia ("Risk Factors for Pneumonia", 2026), so this result is also clinically plausible. In the same vein as our permutation ablation results, performance degradation was small across all single features.

Overall, both methods suggest that the best_EHR model is not overly reliant on any particular feature. This likely contributes to the stability of the model at the cost of limiting interpretabilty of individual predictive features.

Sensitivity Analysis

To evaluate the robustness of the best_EHR model, we conducted a hyperparameter sensitivity analysis focused on max_depth and learning_rate. Our initial hyperparameter optimization strategy for the model used randomized search. This resulted in a tuned model that used a max depth of 8 and a learning rate of 0.01. For our sensitivity analysis, we tested max depths of 7,8, and 9 along with learning rates of 0.005, 0.01, and 0.02.

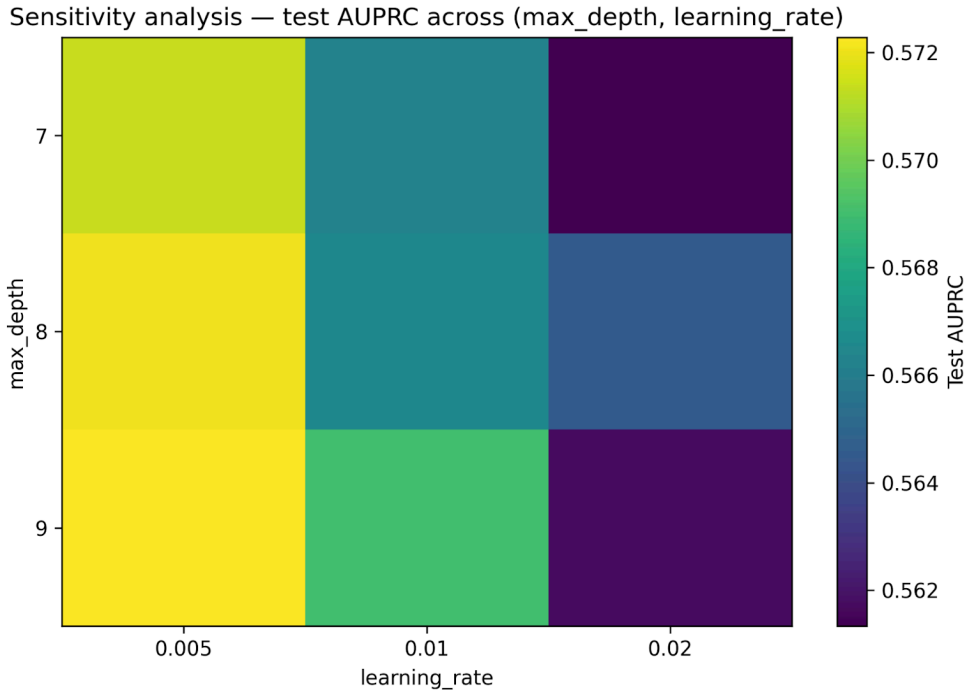


Figure 3: Sensitivity heatmap of test AUPRC across max dept and learning rate

Figure 3 showed that lowering the learning rate to 0.005 improved test AURPCs with the best combination (max_depth=9, learning_rate=0.005) resulting in a test AUPRC of about 0.572. Learning rate appeared to have a more substantial effect on AURPC than max depth, but no configuration resulted in a large degradation in performance. Since all observed AURPC scores were between 0.562 and 0.572, the model appeared to be stable under small changes to hyperparameters. Overall, the sensitivity analysis supports the robustness of the model.

Identify Tradeoffs

In the context of a predictive medical model, threshold selection reflects a fundamental tradeoff between precision and recall. In a clinical setting, a false positive (where the best_EHR model predicts a patient has pneumonia when they do not) will likely trigger further testing and monitoring before treatment. In contrast, a false negative (where the best_EHR model predicts a patient does not have pneumonia when they do) carries the significant risk of delaying treatment, potentially leading to disease progression.

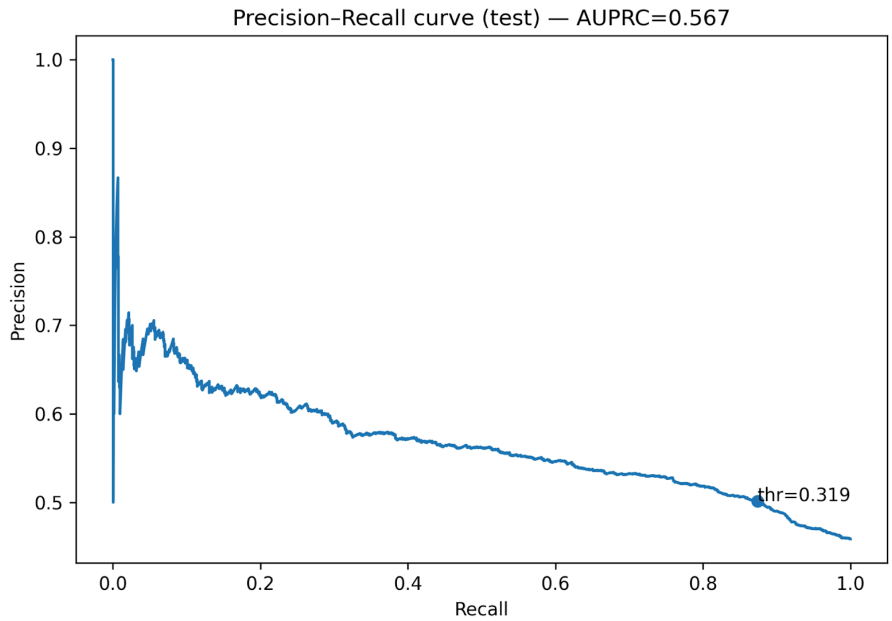


Figure 4: Precision-Recall curve

Figure 4 illustrates how this tradeoff manifests in our model. At the selected operating threshold of 0.319, the model achieves a recall of approximately 0.89 and a precision of approximately 0.50. This indicates that the model captures the majority of true positive pneumonia cases while also generating a large amount of false positives.

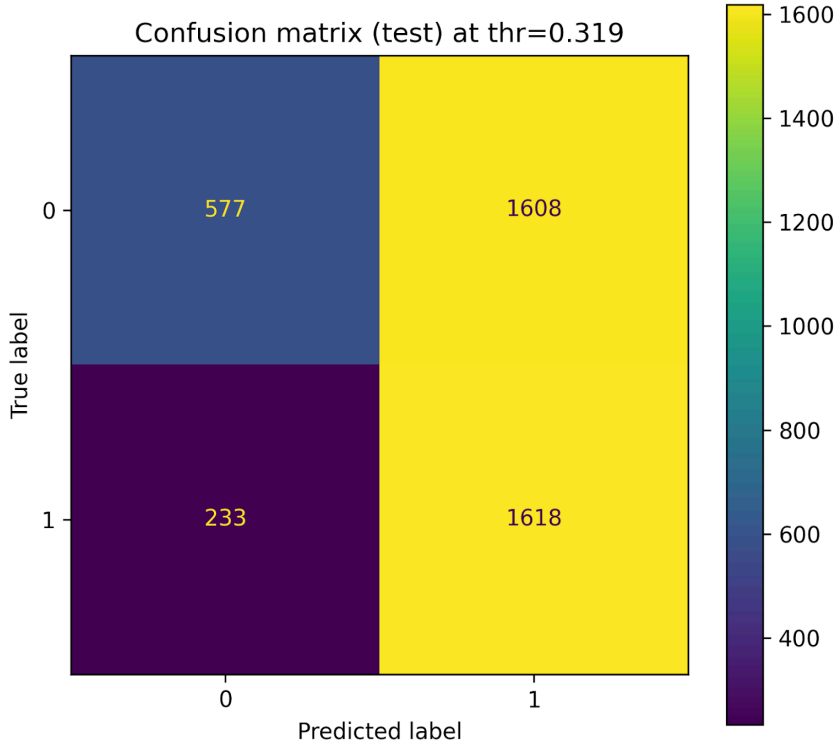


Figure 5: Confusion Matrix at Threshold

Figure 5 shows that there were only 233 false negatives while there were 1618 false positives. This means the model is far more likely to over-flag pneumonia than to miss a real case. Although this reflects a prioritization of minimizing missed diagnoses, it also shows that the model could cause strain in resource-limited environments where extra testing is a burden. More work could be done to improve the model's precision while mitigating the impact on recall.

To better understand the limitations of the best_EHR model, we examined individual test records that were misclassified. We selected examples representing three distinct categories of failure: pneumonia cases with limited lab abnormalities, non-pneumonia cases that were misclassified as pneumonia cases, and sparse records that produced unstable predictions.

Index	Error Type	y_true	y_pred	y_prob	Urea Nitrogen (max)	Anchor Age	Hematocrit (min)	Sodium (max)	Urea Nitrogen (mean)	Hematocrit (max)	Creatinine (mean)	Magnesium (min)	Bicarbonate (max)	Gender
7113	false_negative	1	0	0.14142154	10.0	28	37.5	141.0	10.0	41.1	0.8	1.9	28.0	1
22970	false_positive	0	1	0.9157445	6.0	43	25.4	136.0	5.0	26.5	0.5	—	30.0	1
40327	false_negative	1	0	0.31893152	—	63	—	—	—	—	—	—	—	1

Table 1: Representative Failure Cases

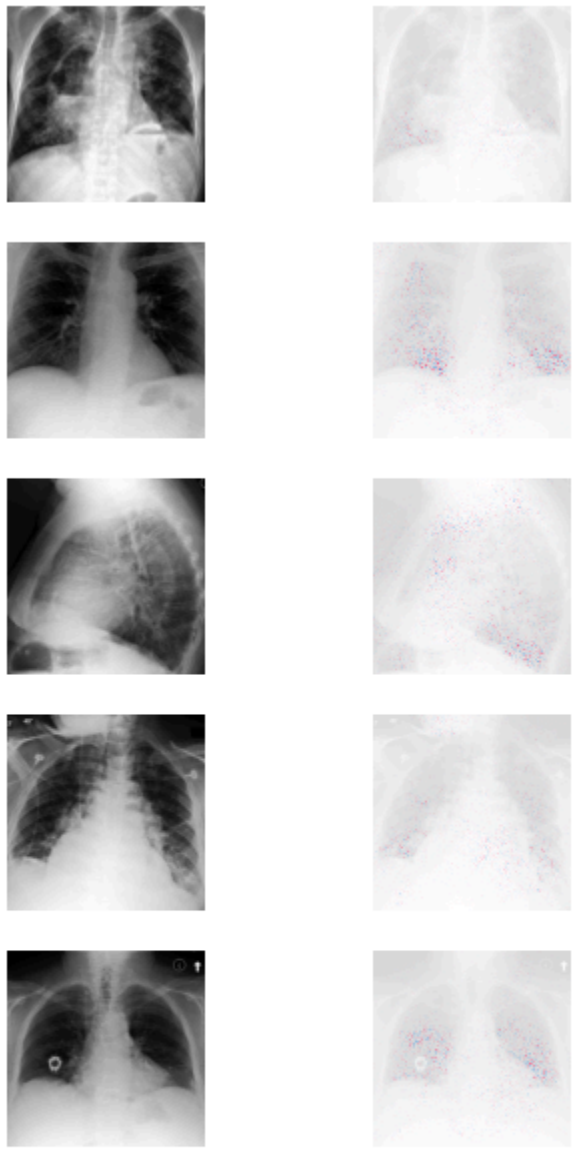
The first category consists of false negatives, where true pneumonia cases were not correctly predicted. For example, record 7113 was a case where the model predicted the patient did not have pneumonia even though they did. However, this is understandable because the patient's lab values are largely within the normal range. It is likely that this is a case of a patient with either an early or mild case of pneumonia. One way of improving the detection of these cases would be to use lab values across a period of time rather than relying on static 24-hour summary statistics.

The second category consists of false positives, where the model predicted patients had pneumonia even though they didn't. For example, record 22970 received a predicted possibility of 0.92 even though the patient did not have pneumonia. The patient has very low hematocrit values, which are associated with more severe cases of pneumonia (Marziano et al., 2025). Cases like these suggest that the model conflates these more general physiological stress markers with pneumonia specifically. Coincidentally, adding imaging data would be a great way to help the model distinguish pneumonia from other conditions. Combining our image-based and EHR-based models is one way we would extend our solution with more time.

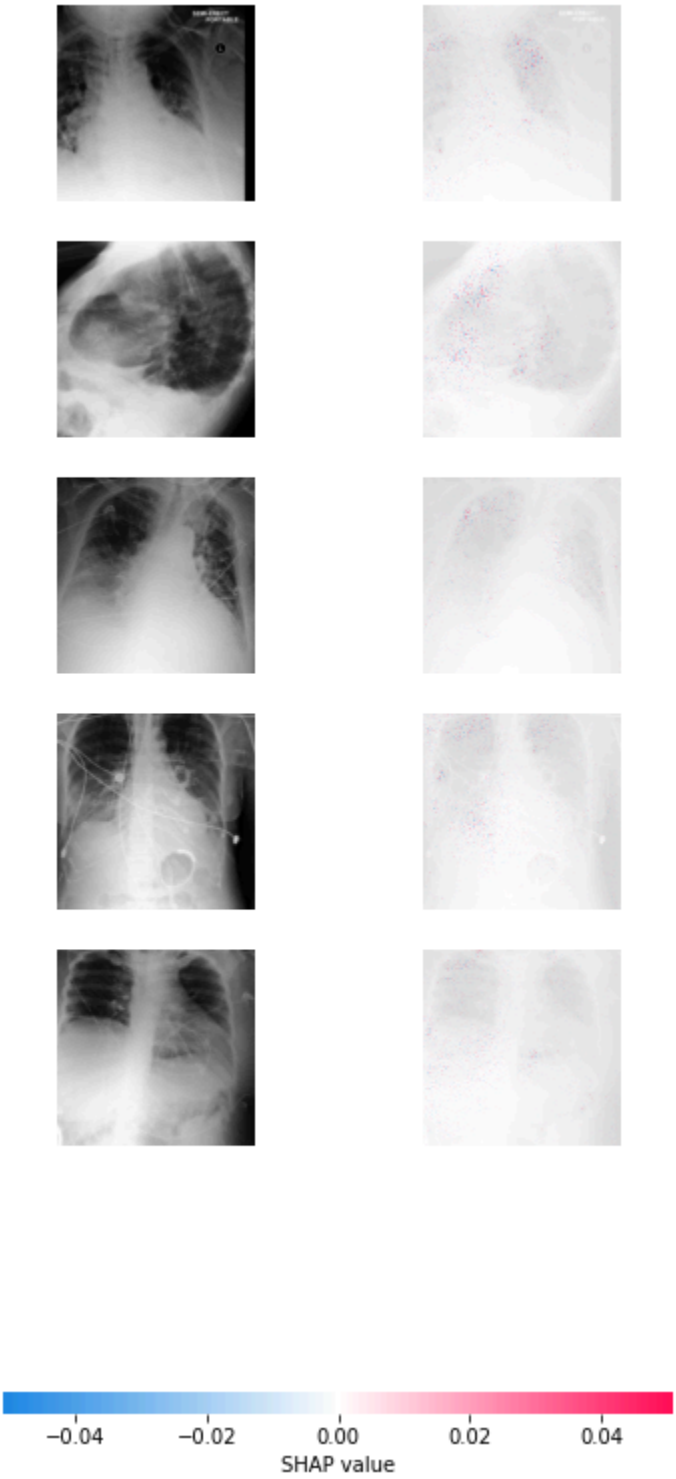
The third category consists of cases like record 40327 where many lab values aren't available. Currently, the best_EHR model is forced to rely on available lab values and demographics because it does not have imaging data available. However, in many cases doctors will order an X-ray to check for pneumonia without running blood work or other labs. This category could also be improved by adding imaging data when available.

Appendix D: Feature Importance (ResNet50_Features.ipynb)

Pneumonia

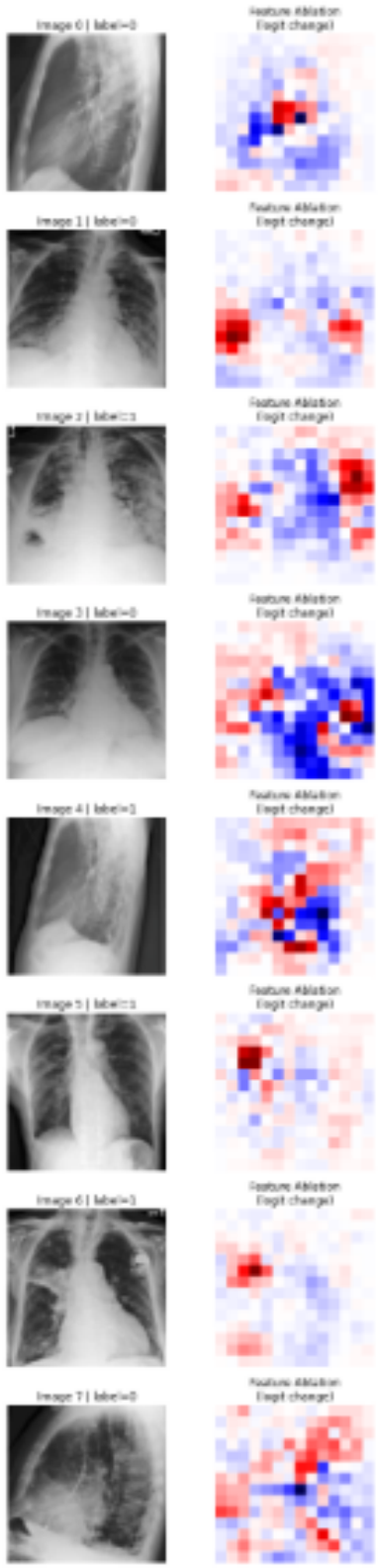


Pneumothorax

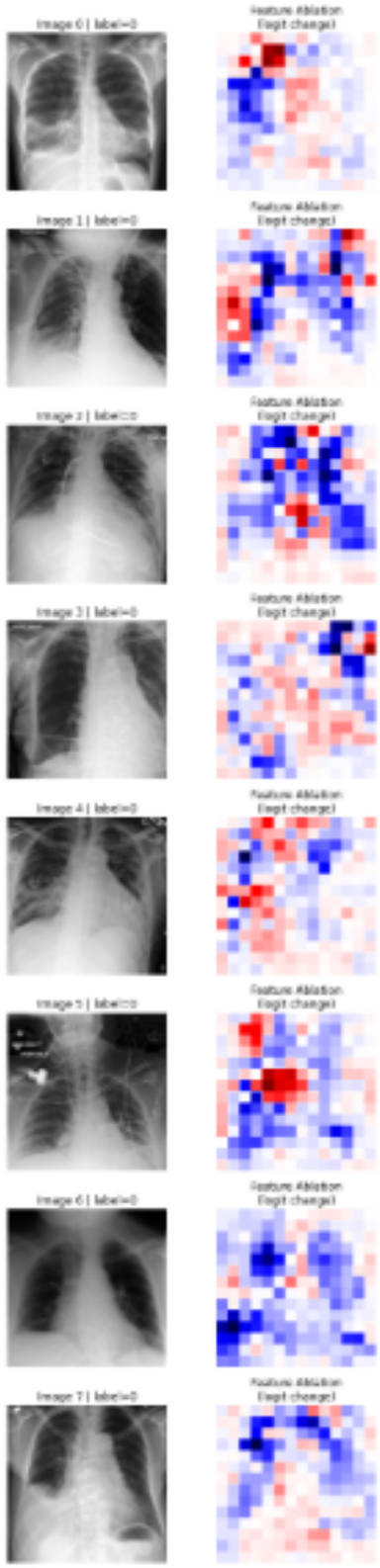


Appendix E: Ablation Analysis (ResNet50_Ablation.ipynb)

Pneumonia



Pneumothorax

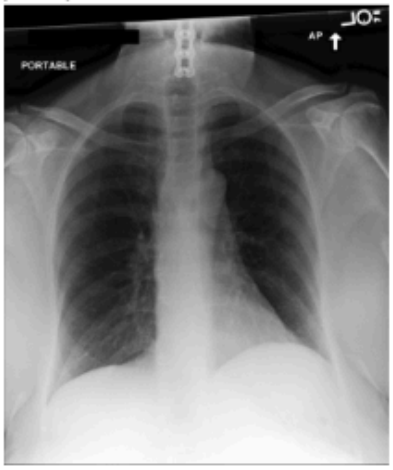


Appendix F: Failure Analysis (ResNet50_FailureAnalysis.ipynb)

Pneumonia

High-Confidence False Positive | true=0 pred=1 prob=0.968 threshold=0.50 5d99dd54-b526167b-ecf0e625-a8664607-ffea3011.jpg

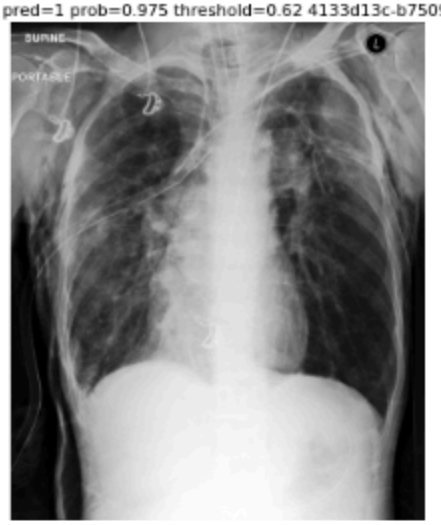
High-Confidence False Negative | true=1 pred=0 prob=0.097 threshold=0.50 bbe6ecaf-aac06564-603fea4c-3e3026e0-8a5cb7c8.jpg



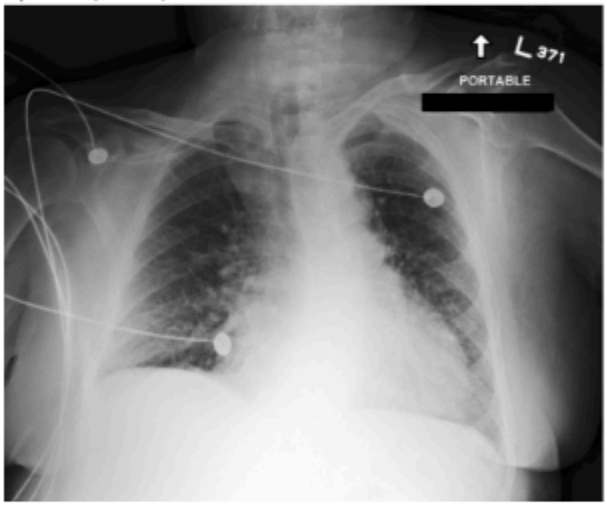
Borderline Misclassification | true=1 pred=0 prob=0.500 threshold=0.50 1a0662d4-8bee75af-c5c452a9-4b43c737-b74d27c1.jpg

Pneumothorax

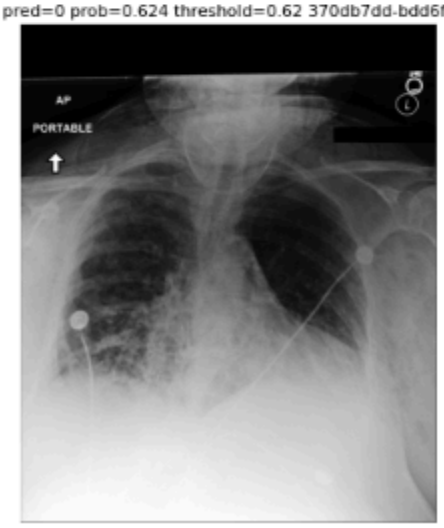
High-Confidence False Positive | true=0 pred=1 prob=0.975 threshold=0.62 4133d13c-b7509603-e76013d9-06a54774-fb044ffb.jpg



High-Confidence False Negative | true=1 pred=0 prob=0.102 threshold=0.62 25fd1806-d10b52d5-9a3103c0-66e21a5f-36fb5086.jpg



Borderline Misclassification | true=1 pred=0 prob=0.624 threshold=0.62 370db7dd-bdd6ffce-5e0e6b83-bc6f534f-61ce5045.jpg



Appendix G: DBSCAN Evaluation Scorecard

eps	min_samples	n_clusters	noise_pct	ari_sample	nmi_sample	silhouette_non_noise	sample_size
23.333333	5	1	0.126832	0.009165	0.001273	NaN	5000
25.555556	10	1	0.074619	0.009150	0.001640	NaN	5000
25.555556	5	1	0.070652	0.008817	0.001607	NaN	5000
25.555556	20	1	0.078916	0.008689	0.001429	NaN	5000
23.333333	10	1	0.133659	0.008292	0.001038	NaN	5000
23.333333	20	1	0.142110	0.007881	0.000931	NaN	5000
21.111111	5	3	0.226847	0.004159	0.000315	0.146286	5000
27.777778	20	1	0.043533	0.003746	0.000469	NaN	5000
21.111111	20	2	0.252271	0.003302	0.000397	0.208234	5000
21.111111	10	1	0.239451	0.003249	0.000211	NaN	5000
27.777778	10	1	0.041190	0.002093	0.000157	NaN	5000
27.777778	5	2	0.038919	0.001735	0.000115	0.281331	5000
12.222222	20	2	0.943806	0.001631	0.001520	0.235356	5000
30.000000	5	1	0.021026	0.001448	0.000135	NaN	5000
30.000000	10	1	0.022190	0.001389	0.000122	NaN	5000
12.222222	5	10	0.920509	0.000940	0.002086	-0.041267	5000
18.888889	20	1	0.416715	0.000643	0.000106	NaN	5000
18.888889	10	2	0.397140	0.000612	0.000218	0.190292	5000
30.000000	20	1	0.023642	0.000549	0.000019	NaN	5000
18.888889	5	3	0.379218	0.000414	0.000177	0.127602	5000